

# Nicolet iS5 FTIR: Routine iD7 ATR SOP

The complete normal-use sequence in one place. Use the OMNIC visual map on the reverse to locate controls.

For users who have completed hands-on FTIR training. This sheet does not replace training or authorize independent use.

## Standard Operating Procedure (SOP)

### 1 Instrument startup

- Power on the instrument, if needed.
- Log on to the workstation using your MIT Kerberos.
- Start **OMNIC**.
- Verify the instrument connection using **System Status**.
- Remove the protective cover from the ATR crystal plate.
- Clean and dry the ATR crystal.
- Collect a background before loading samples.

### 2 Operation

- Wear nitrile gloves when handling samples, ATR accessories, the pressure tower, pressure tips, Kimwipes, or cleaning solvent.
- Remove gloves before using the keyboard, mouse, or workstation.
- Clean the ATR crystal before each sample or whenever contamination is possible.
- Choose the sampling method and pressure tip appropriate for the sample.
- Load the sample so it makes good contact with the ATR crystal.
- Collect the sample spectrum.
- Save each spectrum you need. Spectra must be selected and saved individually.
- Wear gloves again before unloading the sample or cleaning the crystal.
- Repeat background, sample collection, cleaning, and saving as needed.

### 3 Instrument shutdown

- Save all data you need and copy it to your own storage.
- Close OMNIC.
- Log off the workstation.
- Clean and dry the ATR crystal.
- Put the cover on the crystal plate and clamp it in place with the pressure tower.
- **Leave the FTIR powered on.**
- Remove all samples, wipes, and waste.

## The glove rule

**Gloves on:** sample, crystal, tower, tips, wipes, solvent.

**Gloves off:** keyboard, mouse, and screen.

## Before you leave

- Data saved individually and copied
- Crystal and plate clean
- Cover on and clamped
- OMNIC closed; workstation logged off
- FTIR still powered on; area clear

## Stop and ask staff

Uncertain sample, damaged or dirty crystal, failed connection, unfamiliar error, persistent poor signal, or anything outside routine iD7 ATR.



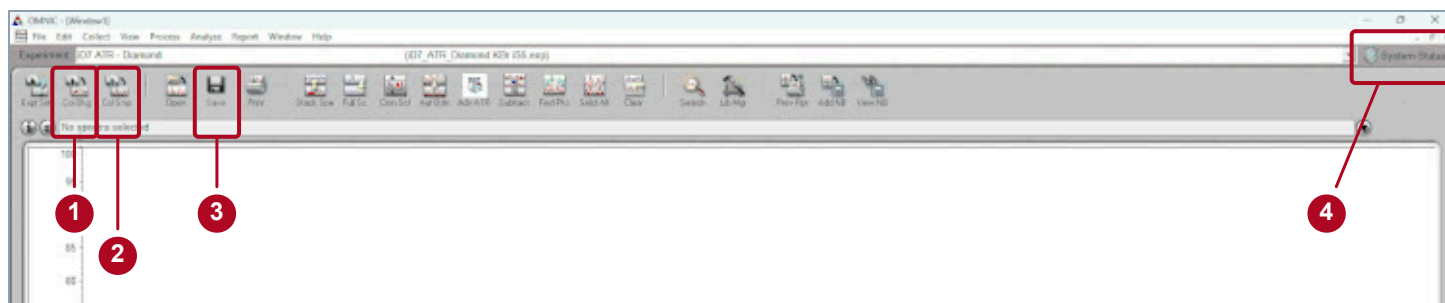
Detailed instructions,  
troubleshooting, and  
help  
[breakerspace.mit.edu/instruments/ftir.html](https://breakerspace.mit.edu/instruments/ftir.html)

# OMNIC visual map

Use this side to find the controls. Follow the SOP on the reverse for the workflow.

## Main OMNIC window

The routine controls stay in the top toolbar.



1 Collect Background

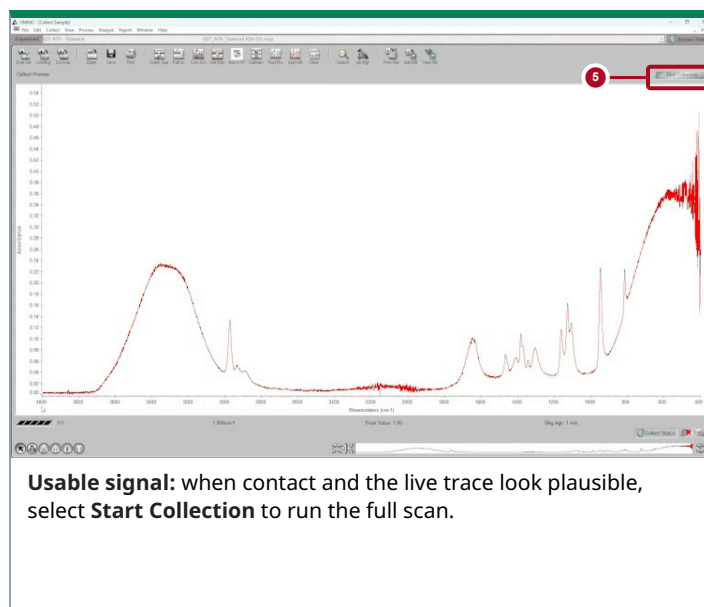
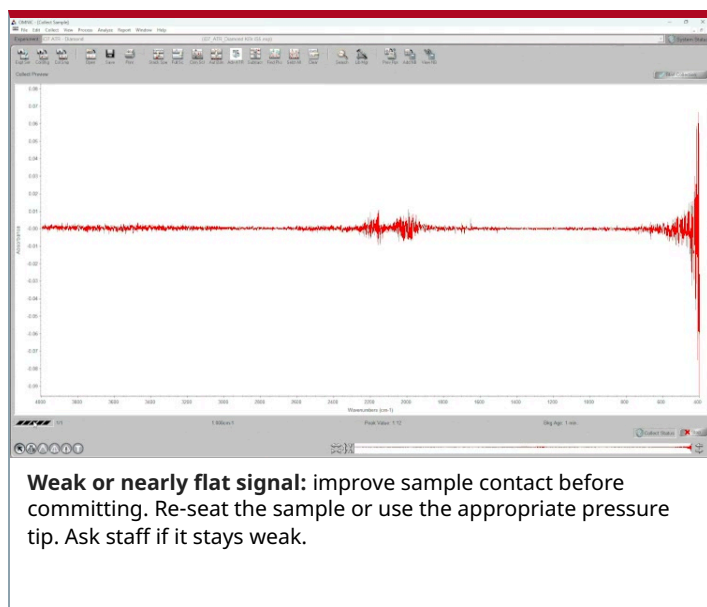
2 Collect Sample

3 Save selected spectrum

4 System Status

## Preview is a decision point

Judge contact before committing the scan. Spectrum shape varies by material; these are illustrative contact states from the recorded FTIR workflow.



## 5 Start Collection

Collect Background or Collect Sample opens the live preview. The scan is not complete until Start Collection runs the full acquisition.

## Before unloading

Check signal and obvious contamination while the sample is still in place. Save each finished spectrum individually.



Full instructions and  
troubleshooting